Reinforcement learning algorithms solve Markov decision processes by estimating value functions or optimizing policies from sampled transitions. Two assumptions underlie the standard formulation. First, the agent's action at time t is determined solely by the observed state s_t and the agent's policy $\pi(a \mid s_t)$; there is no unobserved variable (confounder) simultaneously influencing both the action and the reward or state transition.[1] Second, the observed state s_t is sufficient for prediction, so that conditioning on s_t renders future states independent of past history. Both assumptions are the Markov property restated in causal language. Both fail routinely in economic applications.

The preceding chapters operated under a third assumption that was so natural it required no mention: the analyst controls data collection. In the tabular algorithms of Section **??** and the gridworld study of Section **??**, each method generated its own trajectories by executing actions in a simulator and observing the consequences. The bandit algorithms of Section **??** chose arms and observed payoffs in real time. Even when exploration was limited, the data-generating process was known because the agent's own policy produced it. This chapter drops that assumption entirely. The analyst receives a fixed log of decisions made by someone else, a behavioral policy μ whose functional form may be unknown and whose action choices may depend on variables the analyst cannot observe. The data are observational in the econometric sense: the analyst had no role in generating them and cannot rerun the experiment under a different policy. Identification, not optimization, becomes the central problem.

Unobserved demand shocks that affect both a retailer's pricing algorithm and consumer demand create endogeneity: the observed correlation between price and demand conflates the causal effect with the confounding effect. An RL agent trained on such observational data converges to a biased policy that systematically overestimates the revenue from high prices.

This chapter formalizes the confounded MDP, presents backdoor-adjusted off-policy evaluation as the key algorithmic solution, and demonstrates its practical consequences through a simulation study in a confounded chain MDP. For comprehensive surveys

[1] In causal inference, "outcome" refers specifically to the response variable Y (or the potential outcome $Y(d)$ in the Rubin framework). The RL/MDP framework has no single corresponding concept called "outcome"; the relevant quantities are the reward R_t, the next state S_{t+1}, the return $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k}$, or the value function $V^\pi(s)$. When economists refer to the "outcome model" $\mathbb{E}[Y \mid X, D]$, the RL analogue is the action-value function $Q(s, a)$. Throughout this chapter, I reserve "outcome" for its causal inference meaning and use the specific RL quantity (reward, state, return, value) elsewhere.

of the rapidly growing causal RL literature, including causal representation learning, counterfactual policy optimization, transfer, and fairness, I refer readers to Deng et al. (2023) and da Costa Cunha et al. (2025).

0.1 The Confounded MDP

Standard MDPs assume that the state s_t is fully observed and that the agent's policy depends only on s_t. In the online algorithms of earlier chapters, these assumptions held by construction: the agent executed its own policy in a known environment, so no unobserved variable could simultaneously influence action selection and transitions. When unobserved confounders influence both the behavioral policy and the transitions or rewards, the MDP is confounded. This formalization, developed by Zhang and Bareinboim (2019), Zhang and Bareinboim (2020), and Kallus and Zhou (2020), provides the foundation for causal reasoning in sequential decision problems. The key tool is Pearl's do-operator. The interventional distribution $P(Y \mid \mathrm{do}(X = x))$ is the distribution that arises when X is set externally rather than observed passively, severing all incoming causal influences on X while leaving the remaining data-generating process intact.[2]

Definition D1 (Confounded MDP (Zhang and Bareinboim, 2020))**.** *A confounded MDP is a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{U}, P, r, \gamma)$ where $\mathcal{S}$ is a finite state space, $\mathcal{A}$ is a finite action space, $\mathcal{U}$ is a space of unobserved confounders, $\gamma \in [0, 1)$ is a discount factor, and the dynamics are governed by structural equations:*

$$U_t \sim P_U(\cdot \mid S_t), \tag{1}$$

$$A_t \sim \mu(\cdot \mid S_t, U_t), \tag{2}$$

$$R_t = f_R(S_t, A_t, U_t) + \epsilon_t, \tag{3}$$

$$S_{t+1} \sim f_S(\cdot \mid S_t, A_t, U_t). \tag{4}$$

The behavioral (logging) policy μ depends on the unobserved confounder U_t through Equation (2). An evaluation policy $\pi(a \mid s)$ depends only on the observed state.[3]

[2]The do-operator is formalized within the structural causal model (SCM) framework of Pearl (2009). An SCM specifies endogenous variables $\mathbf{V}$, exogenous variables $\mathbf{U}$, structural equations $V_i = f_i(\mathrm{pa}(V_i), U_i)$, and a distribution $P(\mathbf{U})$. The intervention $\mathrm{do}(X = x)$ replaces the structural equation for X with a constant, producing the interventional distribution. See Pearl (2009) for the complete framework, including the causal hierarchy (association, intervention, counterfactual) and general identification theory.

[3]A confounded MDP differs from a partially observed MDP (POMDP). In a POMDP, the agent knows its

The separation between the behavioral policy μ and the evaluation policy π is the formal signature of offline RL. In the online algorithms of Sections **??**–**??**, the behavior and target policies were identical (on-policy methods) or related by a known exploration mechanism (ϵ-greedy, softmax); here, μ is an unknown function of unobserved variables.

The key distinction is between observational and interventional transition dynamics. Under the behavioral policy μ, which depends on U_t, the conditional distribution $P(s' \mid s, a)$ obtained by conditioning on the observed action does not equal the interventional distribution obtained by setting the action externally:

$$P(s' \mid s, a) \neq P(s' \mid s, \mathrm{do}(a)). \tag{5}$$

The left-hand side conditions on the event $\{A_t = a\}$, which carries information about U_t through the behavioral policy μ. The right-hand side intervenes on A_t, severing the dependence between action and confounder. When U_t also affects S_{t+1} through Equation (4), the conditioning introduces a spurious association between a and s'.

The Bellman equation for policy evaluation must use interventional, not observational, transition probabilities. Define the causal Bellman operator for a target policy π:

Definition D2 (Causal Bellman Operator (Zhang and Bareinboim, 2020))**.** *The causal Bellman operator $\mathcal{T}_c^\pi$ for policy π in a confounded MDP is*

$$(\mathcal{T}_c^\pi V)(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \sum_{s' \in \mathcal{S}} P(s' \mid s, \mathrm{do}(a)) \Big[r(s, a) + \gamma V(s') \Big], \tag{6}$$

where $r(s, a) = \mathbb{E}[R_t \mid S_t = s, \mathrm{do}(A_t = a)]$ is the interventional expected reward.

When an analyst ignores confounding and uses the observational transition probabilities $P(s' \mid s, a)$ in place of $P(s' \mid s, \mathrm{do}(a))$, the resulting value estimates are biased.

Lemma L1 (Bias of Naive Off-Policy Evaluation (Kallus and Zhou, 2020))**.** *Let $\hat{V}^\pi_{naive}$ denote the value function obtained by solving the Bellman equation with observational transitions $P(s' \mid s, a)$, and let V^π denote the true value function under interventional transitions $P(s' \mid s, \mathrm{do}(a))$. In a confounded MDP where $P(s' \mid s, a) \neq P(s' \mid s, \mathrm{do}(a))$*

observation is incomplete and optimizes accordingly; the hidden state affects what the agent sees but the analyst models this explicitly. In a confounded MDP, the hidden variable U_t distorts the behavioral policy μ without the analyst's knowledge or control, so the core challenge is not planning under partial observability but rather identifying interventional quantities from data generated by a policy that depended on variables the analyst cannot observe.

for some (s, a, s'), the naive estimator is biased:

$$\hat{V}^{\pi}_{naive}(s) \neq V^{\pi}(s). \tag{7}$$

The importance-sampling estimator $\hat{V}^{\pi}_{IS} = \frac{1}{N}\sum_{i=1}^{N}\prod_{t=0}^{T}\frac{\pi(a_t^{(i)}|s_t^{(i)})}{\mu(a_t^{(i)}|s_t^{(i)})}G^{(i)}$, where $G^{(i)} = \sum_{t=0}^{T}\gamma^t R_t^{(i)}$ is the discounted return of trajectory i and T is the trajectory length, is also biased because the propensity $\mu(a \mid s)$ is not the true behavioral propensity $\mu(a \mid s, u)$.

The bias arises because the importance-sampling ratio $\pi(a \mid s)/\mu(a \mid s)$ does not correctly reweight the data. The true behavioral policy is $\mu(a \mid s, u)$, but since u is unobserved, the analyst can only estimate the marginal propensity $\mu(a \mid s) = \mathbb{E}_U[\mu(a \mid s, U) \mid s]$. Reweighting by the marginal propensity does not eliminate the confounding bias.[4]

The backdoor criterion (Pearl, 2009) provides a path to identification. Zhang and Bareinboim (2020) apply it to the confounded MDP setting.

Theorem 1 (Backdoor Identification in Confounded MDPs (Pearl, 2009; Zhang and Bareinboim, 2020))**.** *Suppose a set of observed variables $\mathbf{Z}_t$ satisfies the backdoor criterion relative to (A_t, S_{t+1}) in the causal graph of the confounded MDP, meaning that $\mathbf{Z}_t$ blocks all backdoor paths from A_t to S_{t+1} and no element of $\mathbf{Z}_t$ is a descendant of A_t.*[5] *Then the interventional transition probability is identified:*

$$P(s' \mid s, \mathrm{do}(a)) = \sum_{\mathbf{z}} P(s' \mid s, a, \mathbf{z})\, P(\mathbf{z} \mid s). \tag{8}$$

Substituting Equation (8) into the causal Bellman operator (Equation 6) yields an identified, consistent estimator of V^{π}.[6]

[4] This is the sequential analogue of the omitted variable bias in linear regression. In the static case, regressing Y on X without controlling for a confounder U yields a biased coefficient. In the sequential case, the bias propagates through the Bellman recursion and can amplify over the horizon.

[5] A backdoor path from A_t to S_{t+1} is any path in the causal graph that begins with an arrow into A_t (i.e., a non-causal path). In the confounded MDP, $A_t \leftarrow U_t \rightarrow S_{t+1}$ is a backdoor path: U_t causes both A_t and S_{t+1}, creating a spurious association. Blocking all such paths by conditioning on appropriate variables eliminates the confounding bias. See Pearl (2009), Chapter 3.

[6] While Theorem 1 assumes $\mathbf{Z}_t$ blocks all backdoor paths, identification is also possible when $\mathbf{Z}_t$ are merely proxies (effects) of the unobserved confounder. This "proximal" approach to RL employs the theory of negative controls to recover interventional values without satisfying the backdoor criterion (Bennett and Kallus, 2021).

0.2 Backdoor-Adjusted Off-Policy Evaluation

Off-policy evaluation under confounding is precisely an average treatment effect (ATE) estimation problem in a dynamic setting (Bannon et al., 2020). The behavioral policy μ corresponds to the treatment assignment mechanism, the target policy π to the counterfactual treatment regime, and the policy value V^π to the expected potential outcome $\mathbb{E}[Y(d)]$ under the Rubin framework. Importance sampling in OPE corresponds to inverse probability weighting (IPW); doubly robust OPE corresponds to the augmented IPW (AIPW) estimator of Robins et al. (1994). The key challenge, distribution shift between μ and π, is the analogue of the overlap (positivity) condition in causal inference.

In the online setting of earlier chapters, distribution shift did not arise: the data distribution tracked the current policy throughout training, so the state-action frequencies under which value estimates were formed matched the frequencies under which the policy would be deployed.

Theorem 1 yields a concrete estimation procedure. Given logged data $\{(s_t, a_t, z_t, r_t, s_{t+1})\}_{t=1}^N$ collected under behavioral policy μ, where z_t is an observed proxy for the confounder:

1. Estimate the conditional transition model $\hat{P}(s' \mid s, a, z)$ and the proxy distribution $\hat{P}(z \mid s)$ from the logged data.
2. Compute interventional transitions via the backdoor adjustment:

$$\hat{P}(s' \mid s, \mathrm{do}(a)) = \sum_z \hat{P}(s' \mid s, a, z)\, \hat{P}(z \mid s). \tag{9}$$

3. Solve the causal Bellman equation (Definition D2) using the estimated interventional transitions to obtain $\hat{V}^\pi$.

The doubly robust variant combines the fitted action-value function $\hat{Q}(s, a)$ (the RL analogue of the econometric outcome model) with backdoor-adjusted propensities, achieving consistency if either $\hat{Q}$ or the propensity model is correctly specified.

0.3 The Broader Causal RL Landscape

The confounded MDP and backdoor-adjusted OPE address one specific failure mode, but the intersection of causality and reinforcement learning extends considerably further. Three active research directions illustrate the scope.

Causal representation learning for RL seeks state representations $\phi(s)$ that capture

causal mechanisms rather than spurious correlations. When superficial statistics change across environments but the underlying causal structure is preserved, policies learned on invariant causal features generalize where standard representations fail (Schölkopf et al., 2021). da Costa Cunha et al. (2025) formalize this as invariant policy optimization within a multi-environment MDP framework. Sim-to-real robotics illustrates the problem. A robot arm trained in a simulator with blue backgrounds and simple textures may learn to associate visual quirks with correct actions rather than learning the actual object positions that matter. When deployed on a real factory floor with different lighting, the policy fails. Training across multiple visually distinct simulators, same physics but different textures and lighting, forces the state representation to discard renderer-specific features and retain only causally relevant ones like object pose and joint angles (da Costa Cunha et al., 2025).

Counterfactual policy optimization uses structural causal models to generate "what-if" trajectories for credit assignment and data augmentation. Buesing et al. (2019) introduce Gumbel-Max SCMs that produce counterfactual rollouts from a single observed trajectory, enabling off-policy evaluation without importance weights. This follows Pearl's Abduction-Action-Prediction procedure (Pearl, 2009): first infer the exogenous noise U that explains the observed trajectory (abduction), then replace the logged action with the counterfactual action (action), and finally propagate the modified action through the original structural equations to predict the counterfactual outcome (prediction). Oberst and Sontag (2019) extend this to Gumbel-Max SCM-based counterfactual off-policy estimation in healthcare settings, while Forney et al. (2017) demonstrate that counterfactual data-fusion can augment online RL exploration by generating synthetic transitions consistent with the learned causal model. In Atari Breakout, moving the paddle determines whether a brick breaks many timesteps later, but which of the dozens of intervening actions actually mattered? Buesing et al. (2019) answer this by replaying an observed trajectory with one action changed while holding the environment's random outcomes fixed. Because the real and counterfactual trajectories share the same randomness, any difference in reward is caused by that single action change, solving the long-horizon credit assignment problem without the compounding errors of model rollouts or the high variance of importance sampling.

Causal transfer in RL applies the transportability theory of Pearl and Bareinboim

(2014) and Bareinboim and Pearl (2016) to policy transfer across domains. Causal invariance identifies which components of a learned policy remain valid when the environment changes, enabling transfer in settings where standard domain adaptation, which relies on distributional similarity, fails because the intervention targets differ. Autonomous driving illustrates the advantage. A self-driving policy trained in simulation has access to millions of miles of data, but the simulator's graphics differ from real roads. Rather than treating this as one large distribution shift, selection diagrams identify which mechanisms are shared (physical dynamics of steering and braking) and which differ (visual rendering, weather effects) (Bareinboim and Pearl, 2016). The shared mechanisms transfer directly; only the differing components require recalibration with scarce real-world data.

For comprehensive surveys covering these directions and others, including fairness, safety, and partial observability, see Deng et al. (2023) and da Costa Cunha et al. (2025).

0.4 Alternative Identification Strategies

The backdoor criterion requires observing a variable $\mathbf{Z}_t$ that blocks all confounding paths. When no such variable is available, three alternative identification strategies apply, each with a direct econometric analogue.

When the backdoor criterion fails because confounders are unobserved, the front-door criterion offers an alternative if the action A_t affects the next state S_{t+1} only through an observed mediator M_t. The front-door criterion requires: (i) A_t has no direct path to S_{t+1} bypassing M_t, and (ii) no unblocked confounding between M_t and S_{t+1}. When satisfied, a two-stage formula recovers the causal effect: first estimate how A_t affects M_t, then how M_t affects S_{t+1} while adjusting for A_t (Pearl, 2009). This decomposition into direct and indirect effects is the sequential analogue of mediation analysis in econometrics (da Costa Cunha et al., 2025).

The causal RL survey of da Costa Cunha et al. (2025) illustrates the front-door criterion with a mobile wellness intervention. A health app (action A_t) aims to reduce patient cortisol levels (outcome S_{t+1}), but unobserved health consciousness U_t confounds the relationship because health-conscious individuals are both more likely to adopt the app and more likely to have low cortisol regardless. The app affects cortisol only through an observed mediator, supplement adherence M_t. The front-door criterion exploits this structure in two stages. First, the effect of app usage on supplement adherence is un-

confounded because health consciousness does not block the causal path from A_t to M_t. Second, the effect of supplement adherence on cortisol is identified by adjusting for app usage, which satisfies the backdoor criterion for the $M_t \to S_{t+1}$ link. Combining these two estimates recovers the causal effect of the app on cortisol without ever observing health consciousness.

When neither backdoor nor front-door variables are available, instrumental variable methods can identify causal effects. An instrument Z_t must satisfy two conditions: it affects the action A_t (relevance) and its effect on S_{t+1} is channeled entirely through A_t (exclusion restriction). Liao et al. (2024) formalize this as a Confounded MDP with Instrumental Variables (CMDP-IV), where transitions take the form $S_{t+1} = F^*(S_t, A_t) + \epsilon_t$ with unobserved confounders ϵ_t affecting both the behavior policy and transitions. Their IV-aided Value Iteration algorithm recovers the transition function F^* from a conditional moment restriction $\mathbb{E}[F^*(S_t, A_t) \mid Z_t] = \mathbb{E}[S_{t+1} \mid Z_t]$, then runs standard value iteration on the estimated model.

Liao et al. (2024) illustrate with neonatal intensive care unit (NICU) assignment. A hospital must decide whether to admit each newborn to the NICU (action A_t), and this decision is confounded by unobserved severity indicators that affect both the admission decision and the infant's health trajectory. Differential travel time from the hospital to a specialty care provider serves as an instrument Z_t. Relevance holds because longer travel times discourage NICU referrals, shifting the admission probability. The exclusion restriction holds because travel time affects infant health outcomes only through the admission decision, not directly. The conditional moment restriction $\mathbb{E}[F^*(S_t, A_t) \mid Z_t] = \mathbb{E}[S_{t+1} \mid Z_t]$ uses variation in travel time across hospitals to trace out the causal effect of NICU admission on health transitions, recovering the transition function F^* that standard regression conflates with the unobserved confounder.

When confounders are truly latent but proxy variables, noisy correlates of the confounder, are available, proximal causal inference provides identification. Bennett and Kallus (2021) adapt this to sequential settings. The analyst observes two proxies: a treatment-side proxy Z_t (e.g., previous observation O_{t-1}) and an outcome-side proxy W_t (e.g., current observation O_t), both conditionally independent given the latent confounder U_t. Two bridge functions, analogous to inverse propensity scores and Q-functions, are estimated from data by solving conditional moment equations, yielding a doubly robust

estimator of the policy value that is $\sqrt{n}$-consistent without ever observing U_t. Bennett and Kallus (2021) demonstrate with a sepsis management simulator in which physicians choose among fluids, vasopressors, and antibiotics at each decision point. The patient's diabetes status is the latent confounder, partially censored in the medical record so that 20% of diabetic patients appear as non-diabetic in the data. Because the physician observes the true diabetes status when prescribing but the analyst's dataset contains the censored version, the behavioral policy depends on a variable the analyst cannot fully recover. Previous clinical observations O_{t-1} (prior lab values and vitals) serve as the treatment-side proxy Z_t because they correlate with diabetes status and influence the physician's current prescribing. Current clinical observations O_t serve as the outcome-side proxy W_t because they reflect diabetes status and predict future health transitions. In their experiments, the proximal estimator correctly identified which evaluation policy improved over the behavioral policy in 82–100% of test cases, while naive estimators that ignored confounding and standard MDP estimators that assumed full observability both achieved 0% accuracy.

0.5 Simulation Study: Confounded Off-Policy Evaluation

Unlike the simulations in Sections **??**–**??**, where each algorithm generated its own training data through interaction, the estimators here receive a fixed dataset collected under the behavioral policy μ. I construct a 5-state chain MDP $\mathcal{S} = \{0, 1, 2, 3, 4\}$ with two actions (advance, stay) and an absorbing goal at state 4. The causal structure at each step is $Z_t \to U_t \to A_t$ and $Z_t \to S_{t+1}$, where $Z_t \in \{0, 1\}$ is an observed covariate drawn i.i.d. with $P(Z_t{=}1) = 0.5$, and $U_t \in \{0, 1\}$ is an unobserved confounder with $P(U_t{=}1 \mid Z_t{=}1) = 0.9$ and $P(U_t{=}1 \mid Z_t{=}0) = 0.1$. Transitions depend on Z_t, not U_t: the advance action succeeds with probability 0.9 when $Z_t = 1$ and 0.5 when $Z_t = 0$; the stay action is a deterministic self-loop. Rewards are a deterministic step cost $r(s, a) = -1$ for $s < 4$, with $\gamma = 0.9$. The behavioral policy depends on the unobserved U_t: $\mu(\text{advance} \mid s, U_t) = 0.55 + \rho \cdot 0.25 \cdot (2U_t - 1)$, where $\rho \in \{0, 0.2, \ldots, 1.0\}$ controls confounding strength. The target policy always advances. Since Z_t satisfies the backdoor criterion relative to (A_t, S_{t+1}), conditioning on Z_t blocks the confounding path $A_t \leftarrow U_t \leftarrow Z_t \to S_{t+1}$, and Theorem 1 guarantees identification. I compare four estimators: the oracle (analytical Bellman solution using the true interventional transition $P(s'{=}s{+}1 \mid s, \text{do}(\text{advance})) =$

0.70); the naive estimator (Bellman solution using observational transitions $\hat{P}(s' \mid s, a)$ from counts, biased per Lemma L1); the backdoor-adjusted estimator ($\hat{P}(s' \mid s, \mathrm{do}(a)) = \sum_z \hat{P}(s' \mid s, a, z)\, \hat{P}(z \mid s)$, which validates Theorem 1); and a naive importance-sampling estimator using marginal propensities $\hat{\mu}(a \mid s)$, biased per Lemma L1 because the true propensity $\mu(a \mid s, U_t)$ depends on the unobserved U_t. Each configuration uses 2,000 trajectories averaged over 20 seeds.

Table 1 reports bias and root mean squared error across confounding strengths. At $\rho = 0$, all model-based estimators have near-zero bias, confirming correct implementation. As ρ increases, the naive estimator's bias grows monotonically to $+0.30$ at $\rho = 1$: the advance action is more likely when $U_t = 1$ (high effort), which correlates with $Z_t = 1$ (favorable conditions), so observational transitions overestimate the advance success probability ($\hat{P}_{\mathrm{obs}} \approx 0.77$ versus the true 0.70). This per-step bias of $+0.07$ compounds through the Bellman recursion; at $\rho = 1$, the value bias at $s = 0$ ($+0.30$) is nearly three times the bias at $s = 3$ ($+0.11$), confirming the multi-period amplification that the Bellman operator produces. The backdoor-adjusted estimator eliminates this bias (RMSE < 0.03 at all ρ), directly validating the identification formula of Theorem 1. The naive importance-sampling estimator exhibits persistent positive bias across all ρ due to the mismatch between marginal and conditional propensities. Figure 1 displays the bias trajectories.

Table 1: Confounded OPE in a 5-state chain MDP: bias and RMSE of three estimators across confounding strengths ρ (20 seeds, 2,000 trajectories per seed). The unobserved confounder U_t affects the behavioral policy; the observed covariate Z_t affects transitions and satisfies the backdoor criterion.

	Naive		Backdoor		Naive IS	
ρ	Bias (SE)	RMSE	Bias (SE)	RMSE	Bias (SE)	RMSE
0.0	+0.006 (0.003)	0.014	+0.005 (0.003)	0.013	+0.635 (0.087)	0.740
0.2	+0.067 (0.004)	0.069	+0.003 (0.003)	0.015	+0.548 (0.117)	0.748
0.4	+0.125 (0.004)	0.127	+0.000 (0.004)	0.018	+0.654 (0.120)	0.837
0.6	+0.182 (0.003)	0.182	−0.004 (0.004)	0.017	+0.602 (0.093)	0.726
0.8	+0.242 (0.003)	0.242	+0.002 (0.004)	0.019	+0.593 (0.085)	0.699
1.0	+0.300 (0.004)	0.301	+0.008 (0.006)	0.026	+0.589 (0.094)	0.718

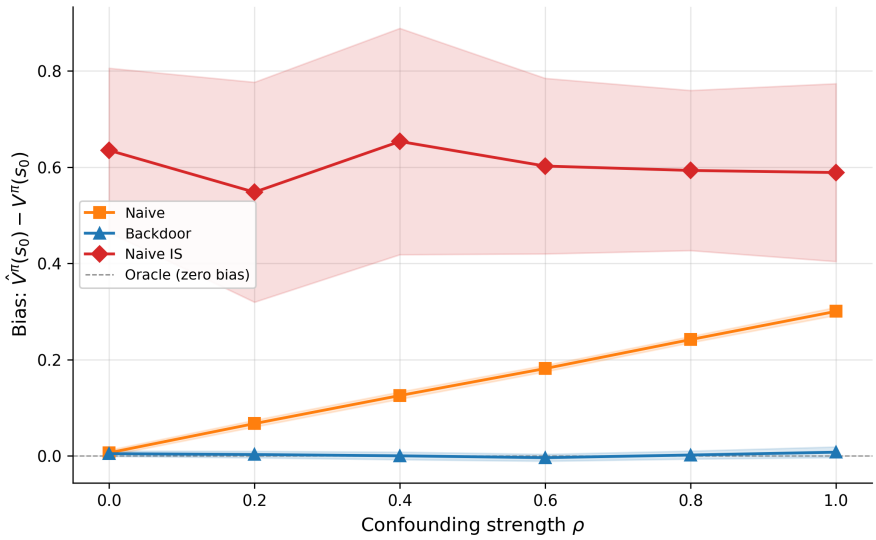

Figure 1: Bias of three OPE estimators relative to the oracle as a function of confounding strength ρ. Naive bias grows monotonically as the behavioral policy's dependence on the unobserved confounder U_t increasingly distorts observational transition estimates. The backdoor-adjusted estimator (Theorem 1) remains unbiased at all confounding levels.

References

James Bannon, Brad Langlois, Raimundo Fernandez, and Danielle Maddix. Causality and batch reinforcement learning: Complementary approaches to planning in unknown domains. In *NeurIPS Workshop on Causal Discovery and Causality-Inspired Machine Learning*, 2020.

Elias Bareinboim and Judea Pearl. Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences*, 113(27):7345–7352, 2016.

Andrew Bennett and Nathan Kallus. Proximal reinforcement learning: Efficient off-policy evaluation in partially observed Markov decision processes. *Operations Research*, 2021.

Lars Buesing, Theophane Weber, Yori Zwols, Sebastien Racaniere, Arthur Guez, Jean-Baptiste Lespiau, and Nicolas Heess. Woulda, coulda, shoulda: Counterfactually-guided policy search. In *International Conference on Learning Representations*, 2019.

Cristiano da Costa Cunha, Wei Liu, Tim French, and Ajmal Mian. Unifying causal reinforcement learning: Survey, taxonomy, algorithms and applications. *arXiv preprint arXiv:2512.18135*, 2025.

Zhihong Deng, Jing Jiang, Guodong Long, and Chengqi Zhang. Causal reinforcement learning: A survey. *Transactions on Machine Learning Research*, 2023.

Andrew Forney, Judea Pearl, and Elias Bareinboim. Counterfactual data-fusion for online reinforcement learners. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1156–1164, 2017.

Nathan Kallus and Angela Zhou. Confounding-robust policy evaluation in infinite-horizon reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

Luofeng Liao, Zuyue Fu, Zhuoran Yang, Yixin Wang, Dingli Ma, Mladen Kolar, and Zhaoran Wang. Instrumental variable value iteration for causal offline reinforcement learning. *Journal of Machine Learning Research*, 25(303):1–56, 2024.

Michael Oberst and David Sontag. Counterfactual off-policy evaluation with Gumbel-Max structural causal models. In *Proceedings of the 36th International Conference on Machine Learning*, pages 4923–4932, 2019.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2nd edition, 2009.

Judea Pearl and Elias Bareinboim. External validity: From do-calculus to transportability across populations. *Statistical Science*, 29(4):579–595, 2014.

James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866, 1994.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021.

Junzhe Zhang and Elias Bareinboim. Near-optimal reinforcement learning in dynamic treatment regimes. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Junzhe Zhang and Elias Bareinboim. Designing optimal dynamic treatment regimes: A causal reinforcement learning approach. In *Proceedings of the 37th International Conference on Machine Learning*, pages 11012–11022, 2020.